

AES Installation Instructions

Designers and Installers must successfully pass the online training course prior to submitting designs or installing AES systems.



AES (Advanced Enviro-Septic™) Wastewater Treatment

Distributed by Environment Technology

For more information and full installation support:

Tel: (64) 3 970 7979 or 0800 waste H20

info@et.nz

www.et.nz



1. Site Preparation & Installation



2. Site Preparation

- 🌐 Remove all organics and sod from areas to receive System Sand
- 🌐 Rip or scarify receiving soil prior to placement of System Sand
- 🌐 Place System Sand immediately after excavating the receiving area
- 🌐 Divert surface water with subtle swales from running into or over system during construction
- 🌐 Do not work wet or frozen soils

3. Couplings



1. Pull fabric back to expose at least two corrugations.



2 . Place coupling under pipe capturing two corrugations on each pipe (if possible).



3. Clamp coupling

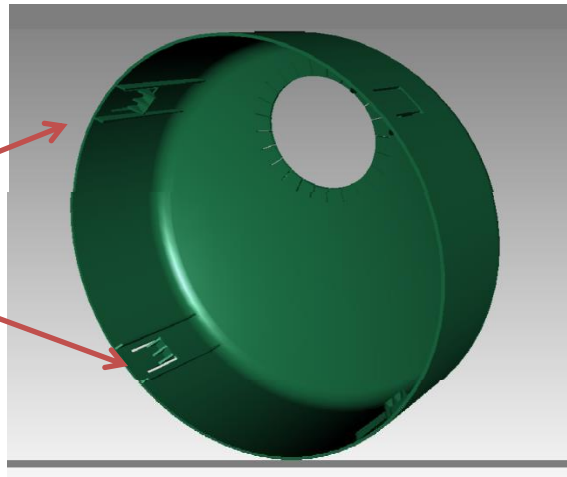
4. Pull fabric over edges of coupling

4. Offset Adapter

1. Pull fabric back to expose corrugations.
2. Place offset adapter over end of pipe
3. Push until the locking tabs engage corrugations.
4. Orient hole at the 12 o'clock position – in line with seam
5. Pull fabric over edge of adapter



Locking
Tabs



5. Cutting AES Pipe

1. Cut fabric with sharp utility knife or commercial grade shears



2. Use saw or utility knife to cut pipe.



6. Pipe orientation

Geo-textile fabric seam (12 o'clock) position

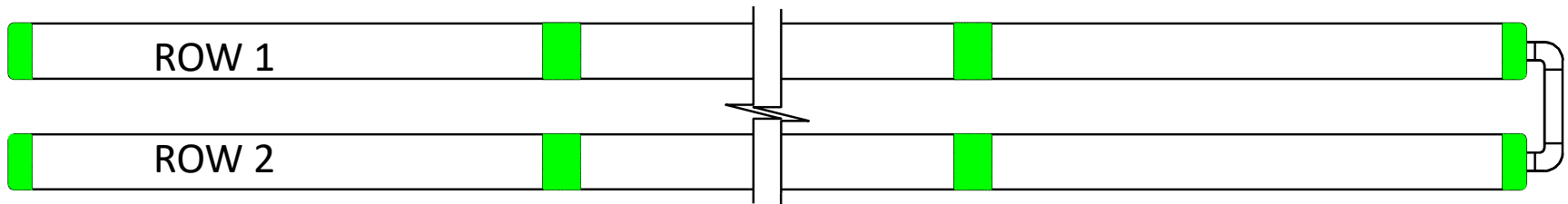
Bio-Accelerator™ fabric (6 o'clock) position



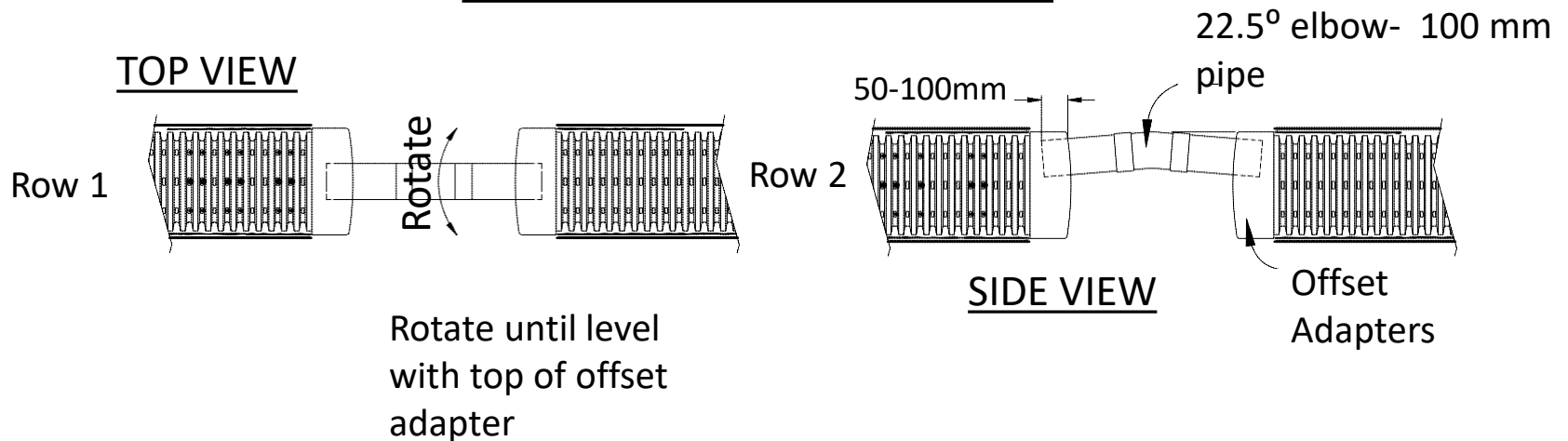
7. Rules for Rows

- 30m maximum row length per bed
- 90 LPD per AES pipe

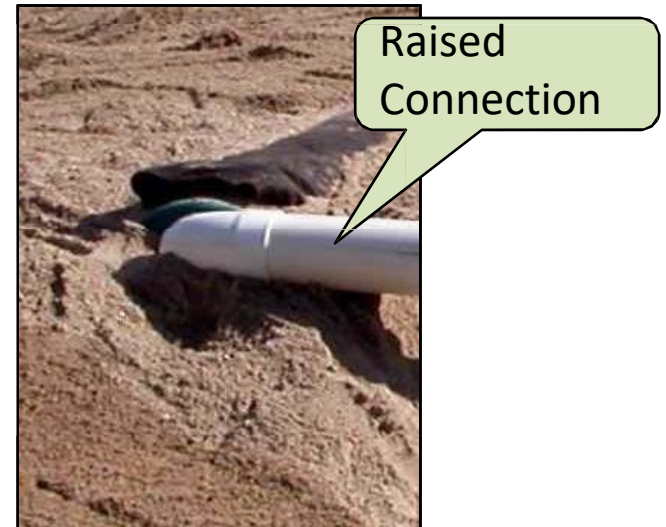
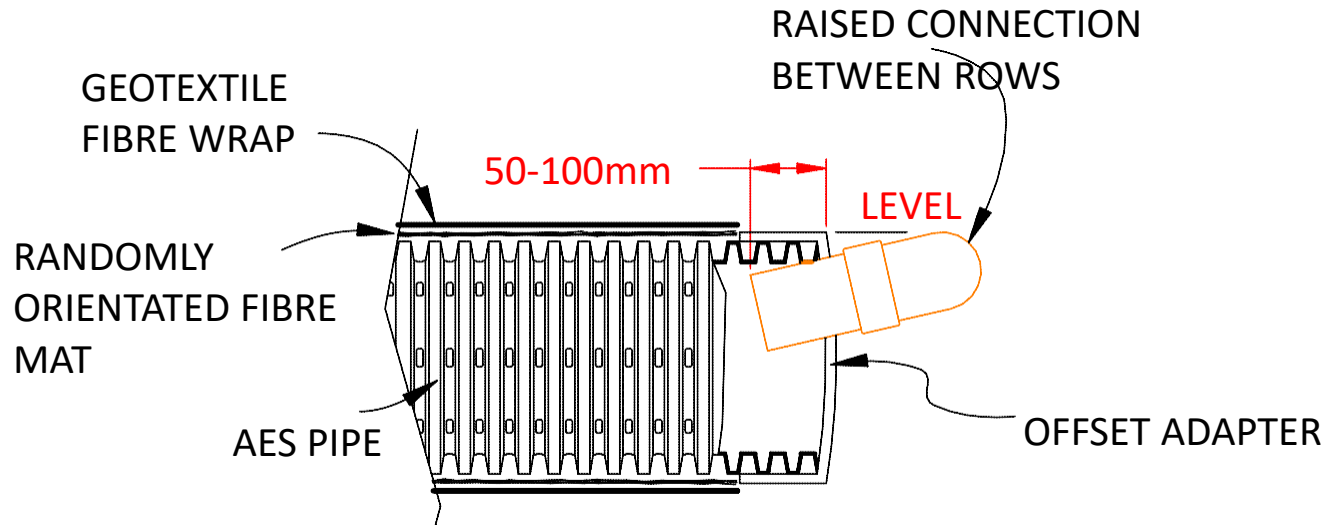
RAISED CONNECTION BETWEEN PARALLEL ROWS



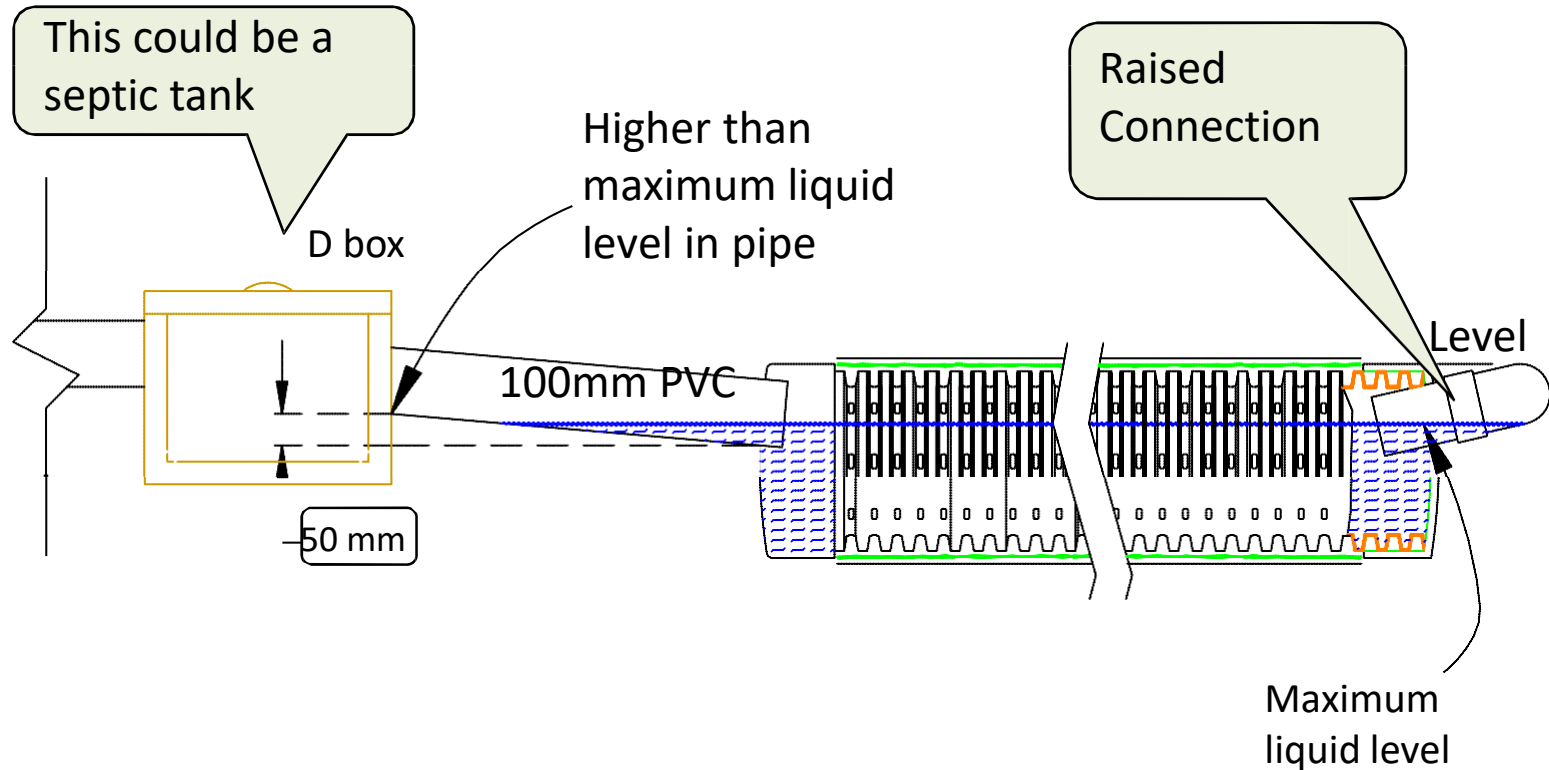
RAISED STRAIGHT CONNECTION



8. Raised Connection Assembly

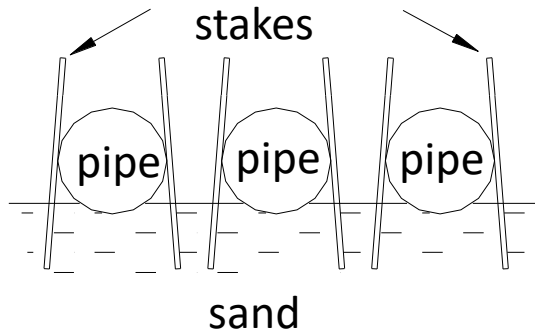


9. Minimum Drop To Pipe – the 50mm rule



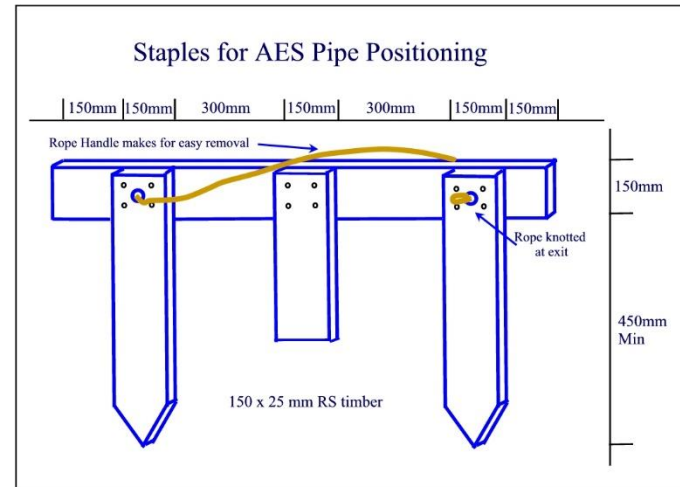
There must be a minimum 50mm drop in elevation to the AES pipe

10. Row spacing

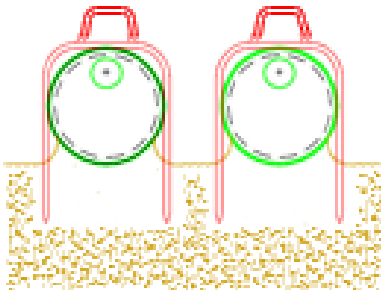


Use stakes to hold pipe during backfill

11. Row spacing



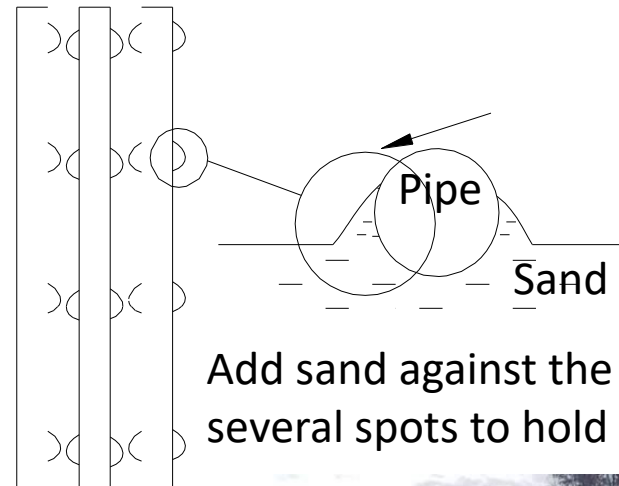
Rebar Frame



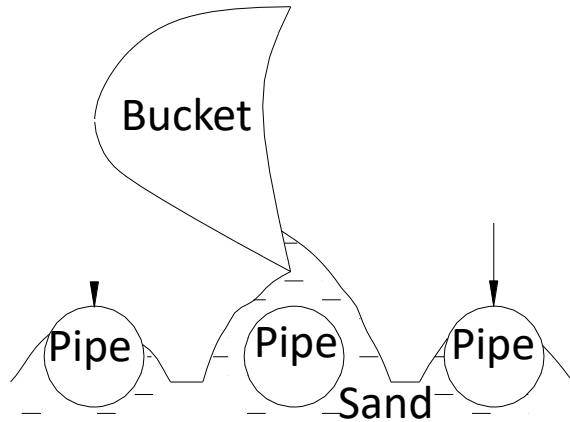
Berm sand at
base of AES pipes
to hold in place



12. Backfilling



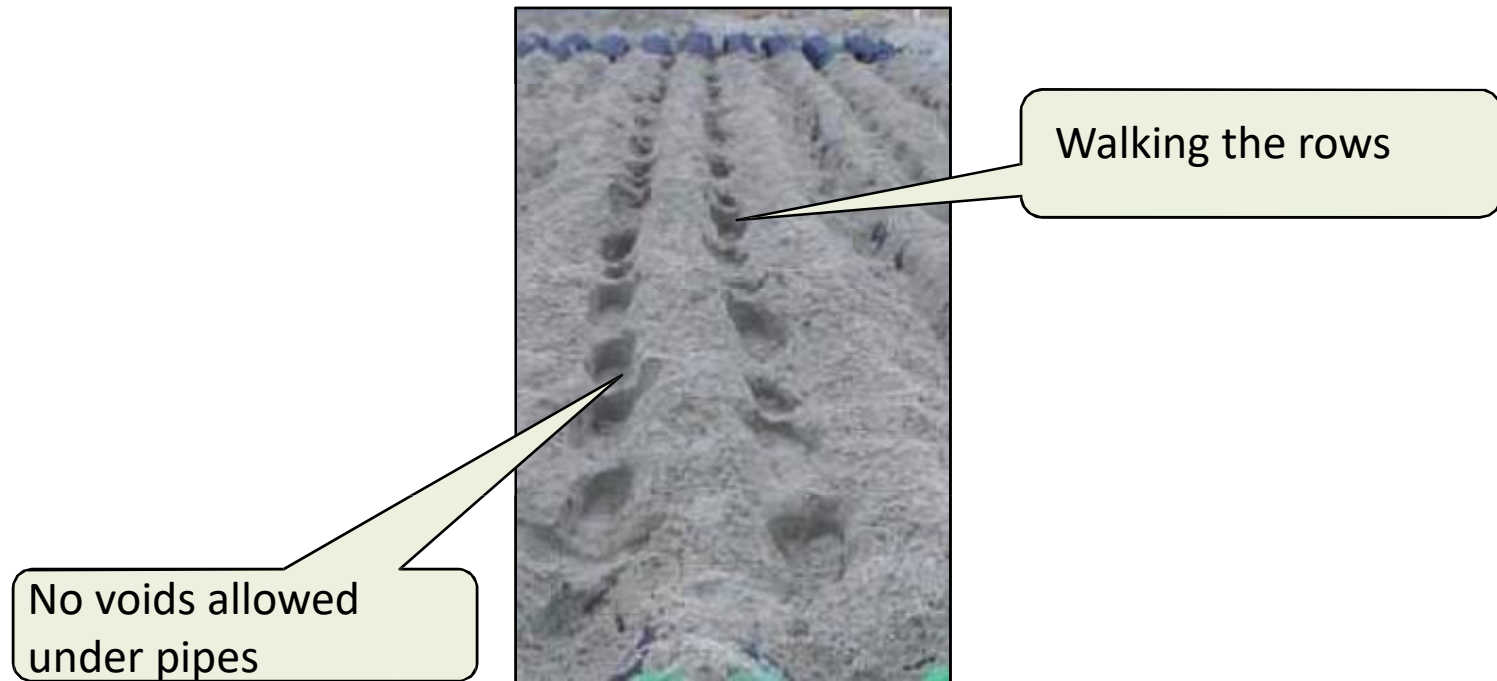
Add sand against the pipe in several spots to hold pipe in place.



Sprinkle a little sand on top of each pipe to hold in place then backfill between them



13. Eliminate sand voids



- 🌐 No mechanical compaction allowed
- 🌐 150mm minimum of System Sand over AES pipes
- 🌐 150mm of topsoil over System Sand

AES INSTALLATION INSTRUCTIONS

AES components are supplied by Environment Technology, 03 970 7979, info@et.nz.
Before purchasing AES components all drainlayers must complete the AES Certification course:
(online at <https://www.et.nz/aes-online-course/>)
An Owner's Manual accompanies every order of AES components

AS/NZS 1547:2012 L7.1 Good construction techniques p148:

The following excavation techniques shall be observed so as to minimise the risk of damage to the permeability of the soil:

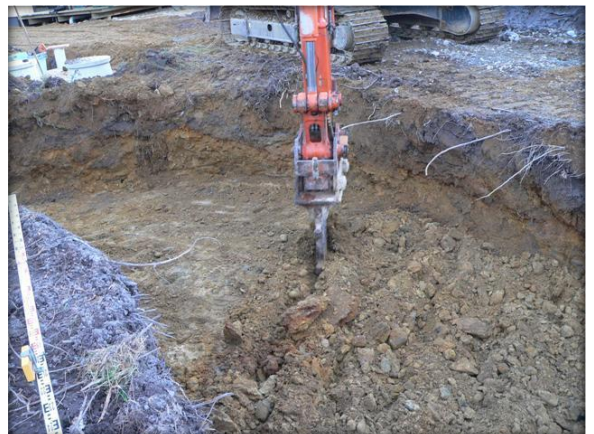
- a) Excavate only when the weather is fine;
- b) Avoid excavation when the soil has a moisture content above the plastic limit. This can be tested by seeing if the soil forms a 'wire' when rolled between the palms;
- c) When excavating by machine, use a bucket with sharp teeth and excavate in smaller 'bites' to minimise compaction;
- d) During the 'wet' season or when construction cannot be delayed until weather becomes fine, smeared soil surfaces may be raked to reinstate a more natural soil surface, taking care to use fine tines and only at the surface;
- e) Avoid compaction by keeping people off the finished trench or bed floor.
- f) In particular for trenches and beds:
 - i) If rain is forecast then cover any open trenches, to protect them from rain damage;
 - ii) Excavate perpendicular to the line of fall or parallel to the contour of sloping ground; and
 - iii) Ensure that the base of the excavation is horizontal parallel to the slope

1. Set out

- a) Ensure that the set out of the work is in accordance with the design approved by Council.
- b) Each row of AES System pipe must be horizontal from end to end.
- c) The AES calculator footprint dimensions are based upon the DLR of the receiving soil and the influent volumes to the AES system. They are the **minimum** foot print area of the AES bed.
- d) Any system extension must be to the down slope side of the AES installation **unless** the infiltration footprint is horizontal or level.

2. Excavation – (track machinery causes less compaction of the soil.)

- a) A laser level and receiver which has been checked for calibration is a necessity for accurate bed excavation and the placement of system sand under the AES pipe.
- b) Strip vegetation from the bed and system extension area including the area which will be covered by the constructed raised bed and batters if a raised bed is required.
- c) Strip topsoil to a stockpile and keep separate for re-covering the installation.
- d) Excavate to reduce levels to a minimum of 475mm below the effluent inlet invert level at the AES bed, leaving the base of excavation loose to aid infiltration. This 475mm is made up of 175mm to AES pipe invert from inlet pipe invert plus 300mm of system sand below the AES pipe.
- e) Ensure the sub-base of the bed is left in this loose and friable state by ripping or scarifying the infiltration area to a minimum depth of 150mm, parallel to the AES pipe, particularly in Category 4,5,6 soil with high clay content.

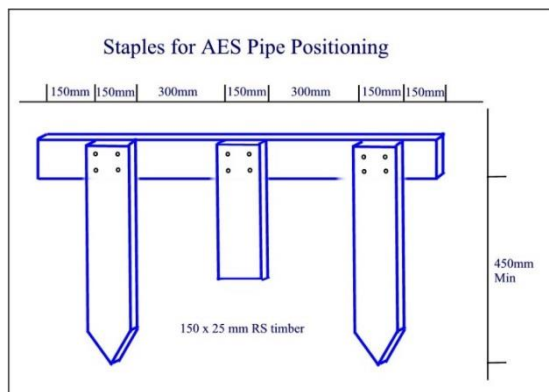
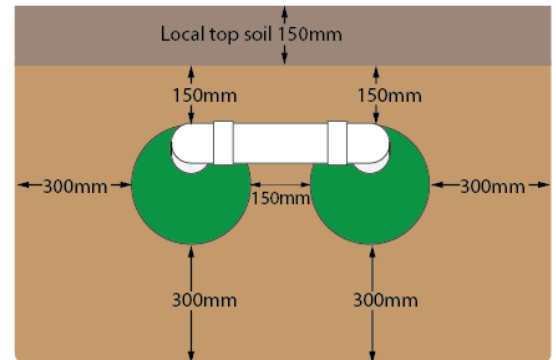


- f) If there is a raised bed configuration required ensure you have sufficient material on site including finishing topsoil to create and cover the entire raised bed or bring in enough sand to fill out batters (1 vertical: 3 horizontal) prior to covering with topsoil etc.
- g) Experience shows that it is simpler to create a mound out of site or imported soils **and then** excavate into the mound to form the AES bed.

3. System Sand – Coarse washed sand

System sand is coarse washed sand (>40% 0.5mm-2.0mm, with <2% silt). Sand can be tested free of charge by ET. Send 2 cups to 14 Onekaka Ironworks Road, Takaka 7182.

- a) Place a minimum of 150mm of system sand in the bed extension area and then a minimum 300mm under the actual AES pipe footprint area. Ensure this thickness of sand under each row of AES pipe has a completely level surface and is evenly compacted by foot traffic.
- b) Place rows of AES pipe roughly in position with the fabric seam at the top or 12 o'clock position and the white bio-accelerator fabric at the bottom or 6 o'clock position. Leave a 300mm minimum clearance to all of the footprint edges.
- c) Join lengths of AES pipe with AES couplings. To do this slide the fabric and the fibre back on the two pipe ends to be joined. Clip the AES coupling in place. Slide the fibre and fabric layers back over the coupling. Ensure that the base of the excavation is horizontal.
- d) Place offset adaptors at the beginning and end of each row of AES pipe with the 100mm hole at the top.
- e) Ensure there is a minimum separation of 150mm between rows of AES pipes. This can be done with pegs, short pieces of 150mm pvc or preferably using AES 'staples' constructed from 150mmx25mm timber. One side must be set to provide the minimum 300mm of system sand required at the footprint edge
- f) Place system sand around AES pipes ensuring they stay level and in position. Pipes must be restrained while sand is being placed. This is most easily achieved with the wooden staples. Remove and progressively position staples or other spacers until all pipes are surrounded by system sand to the top or soffit of the pipes. Walk on the sand between rows to aid compaction with the first compaction when the sand is 2/3rds the thickness of the pipes.



4. Connecting Rows

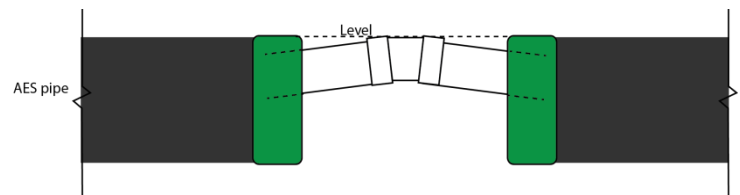
- Connect rows with 100mm pipe as required with a maximum 100mm extending through the Off-Set Adaptor into the AES system pipe.
- Raised connections: After placing raised connection pipes, the top of the PVC must be level with the top of the AES pipe. Lift and pack with compacted sand. This ensures the airflow is not restricted even when the AES pipe is working to maximum buffer capacity. See diagram overleaf.



Offset adaptor



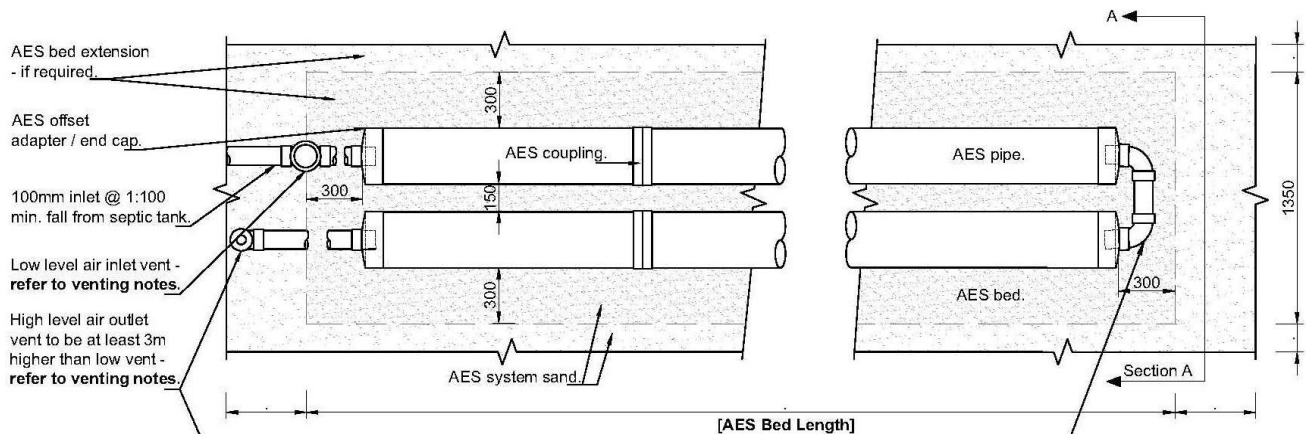
Raised connector



In-line raised connector

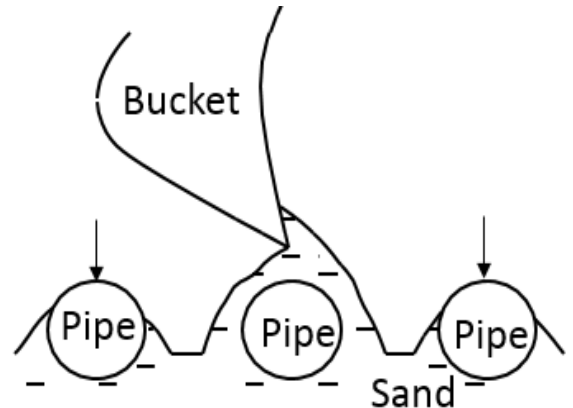
5. Venting

- You must ensure the system has a high/air outlet vent and a low/air entry vent as shown in the design. The Low vent must be a minimum of 150mm above ground preferably in a protected position away from mowers or vehicles, and above snowfall level. Vents can be located any distance from the system provide they have no water traps where condensation could collect. The High Vent must be a minimum of 3 meters higher than the low vent.
- Collect 'As Built' information by recording the position of all AES pipe end points and any other required specific features and inspection points between the septic tank and the AES bed.
- Usually rows are connected in series with one low vent as pictured following:



6. Back Filling

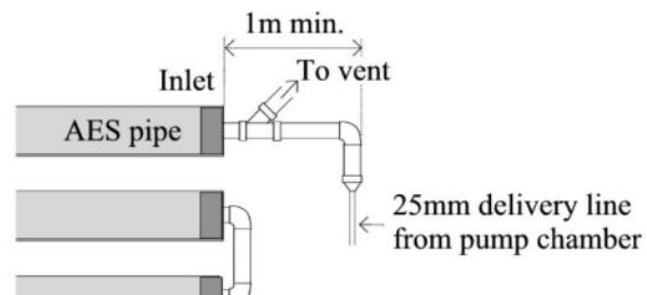
- a) Ensure a minimum of 150mm of System Sand covers the AES pipes and connecting PVC pipe work.
- b) Refer to the onsite design and Council approval and ensure that all stormwater diversions, drains or site specific requirements are correctly installed.
- c) Backfill with natural soil and compact. System extensions may require compaction in multiple layers depending on the depth of backfill material to be placed.
- d) On raised beds and down slopes strip vegetation and place fill evenly and level to all sides to avoid breakout from low points during high seasonal loadings.
- e) Cover excavation area with topsoil creating a finished surface level 50 to 100mm higher than the natural surface level ensuring that water sheds off the land application area and does not pond, compact lightly and seed or grass when completed.



7. Use of Velocity Diffuser

Pressurised or steep gravity systems require a **Velocity Diffuser** - a 100mm 90° bend 1m before the AES pipe inlet – to minimise turbulence when the septic tank effluent enters the AES pipes.

A distribution box can also function as a velocity diffuser.



8. Use of Filter Cloth/ Geo-Fabric

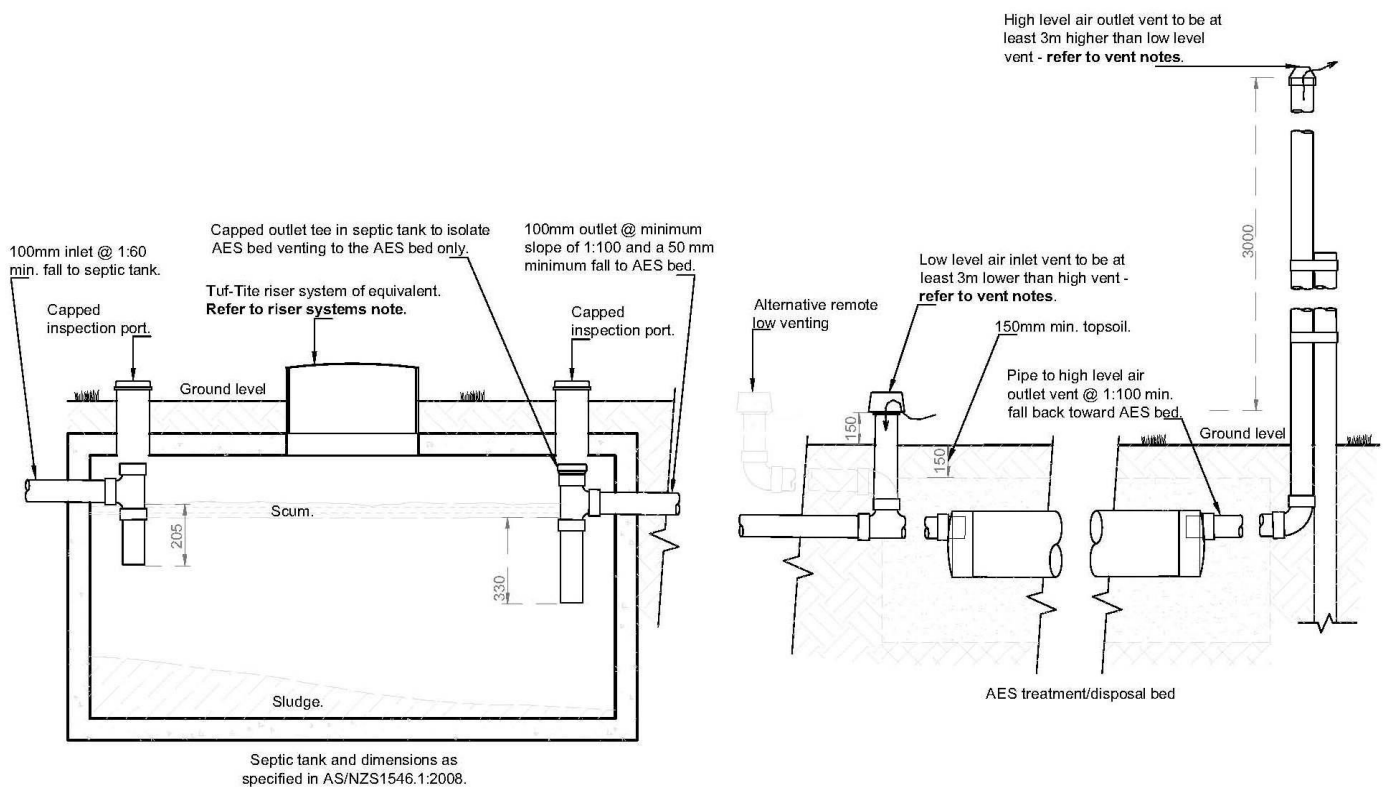
The manufacturer of AES strongly advises against using geo-fabric between the topsoil and the AES sand, as it can become clogged and prevent gasses from escaping. It is up to the local designer or installer whether geo-fabric should be used or not.

**For supply of AES components, sand specification testing and installation support phone
Environment Technology 03 9707979, 0800 Waste H2O (927 834) email: info@et.nz
14 Onekaka Iron Works Rd, Takaka 7182 and 105 Pascoe St, Tahunanui, Nelson 7011**

Venting & Septic Tanks

It is crucial for there to be passive air flow through the whole AES bed. The advanced secondary treatment that takes place in an AES bed is due to aerobic conditions being maintained and the provision of habitat for bacteria to thrive in. This is achieved by having a 100mm Low Vent at one end of the AES bed, and a High Vent at the other end of the bed that is at least 3m higher than the low vent. The vents can be positioned either at the bed or remotely/ at a distance from the AES bed if this is more practical or for aesthetic reasons.

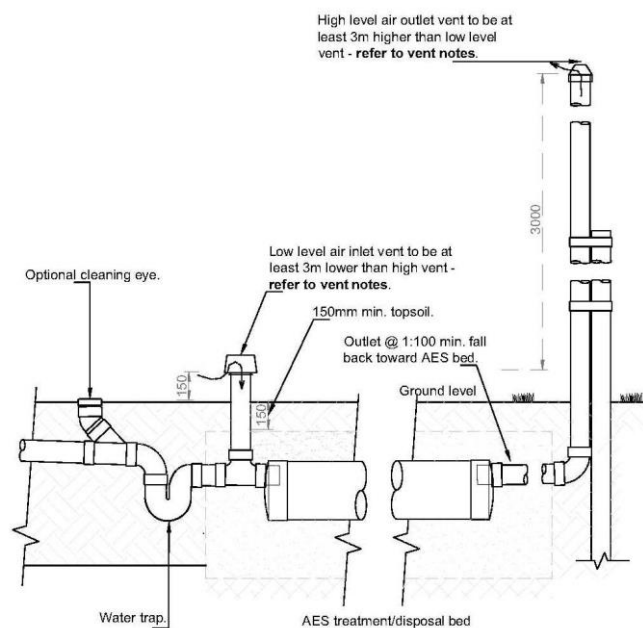
Septic tank set-up that allows free passage of air:



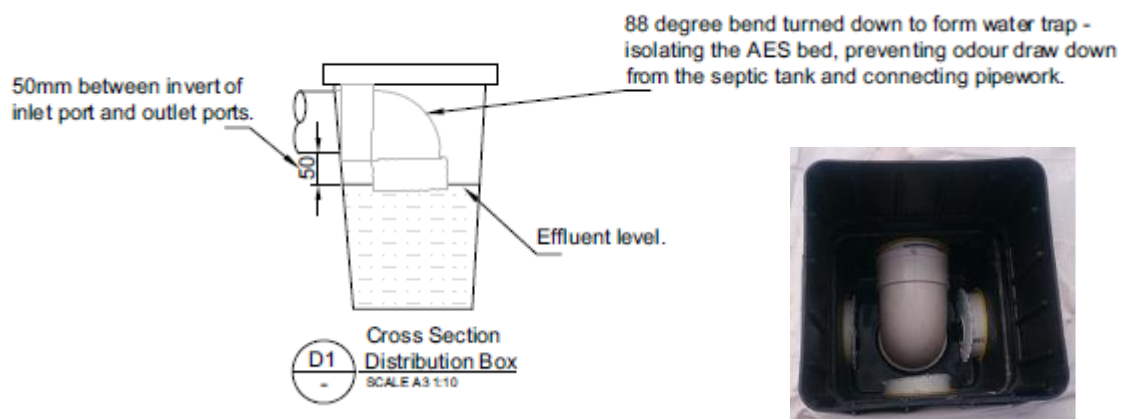
Maintenance

All vents are to be kept free of vegetation.

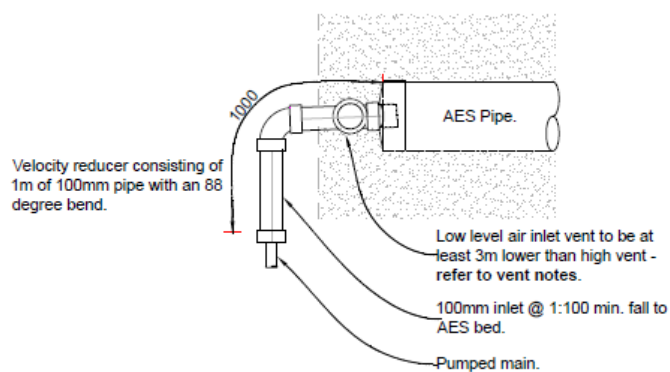
Venting with a water trap:



Venting with a Distribution Box



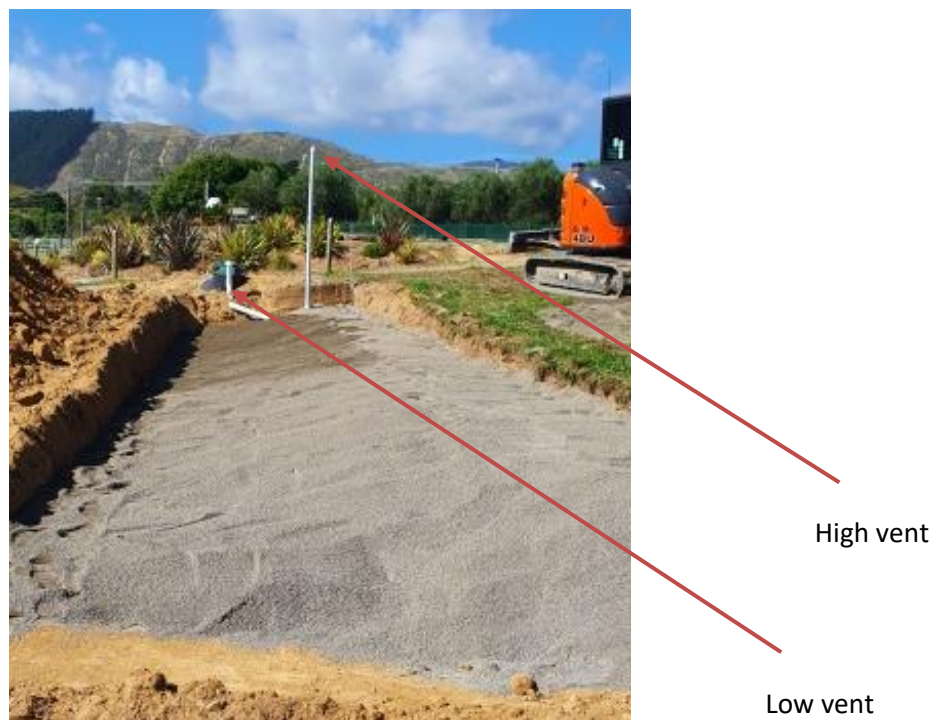
Venting with a velocity reducer:



The low vent is always a minimum of 100mm in diameter, and installed at a minimum 150mm above ground level, or above possible snowfall depth. If the High Vent is supported by a building, the vent should be positioned above roof height and away from windows and as required for terminal vents in G13 5.



In this system the septic tank effluent is pumped to a raised AES bed. Both high and low vents are remote.



Any low points in the venting pipework must have holes to allow water (condensation/ rain) to drain.

The low point in the vent pipework pictured has holes drilled in the bottom of the low point, and is surrounded by drainage metal wrapped in filter cloth.

AES System Sand Requirements

The AES system sand is coarse filter sand, closely aligned with C33 sand specifications. The primary function of the AES system sand is to transport/allow movement of liquid and air to provide an environment around the AES pipes where oxygen dependant microorganisms can continue to thrive. If the sand is too coarse, there will be no capillary action and the effluent will drain through too rapidly, if too fine will become impermeable/anaerobic and drain through too slowly. If the silt content is above 2%, the silt migrates to the base of the bed and the soil can become less permeable, meaning the effluent disposal field has the potential to bind and fail.

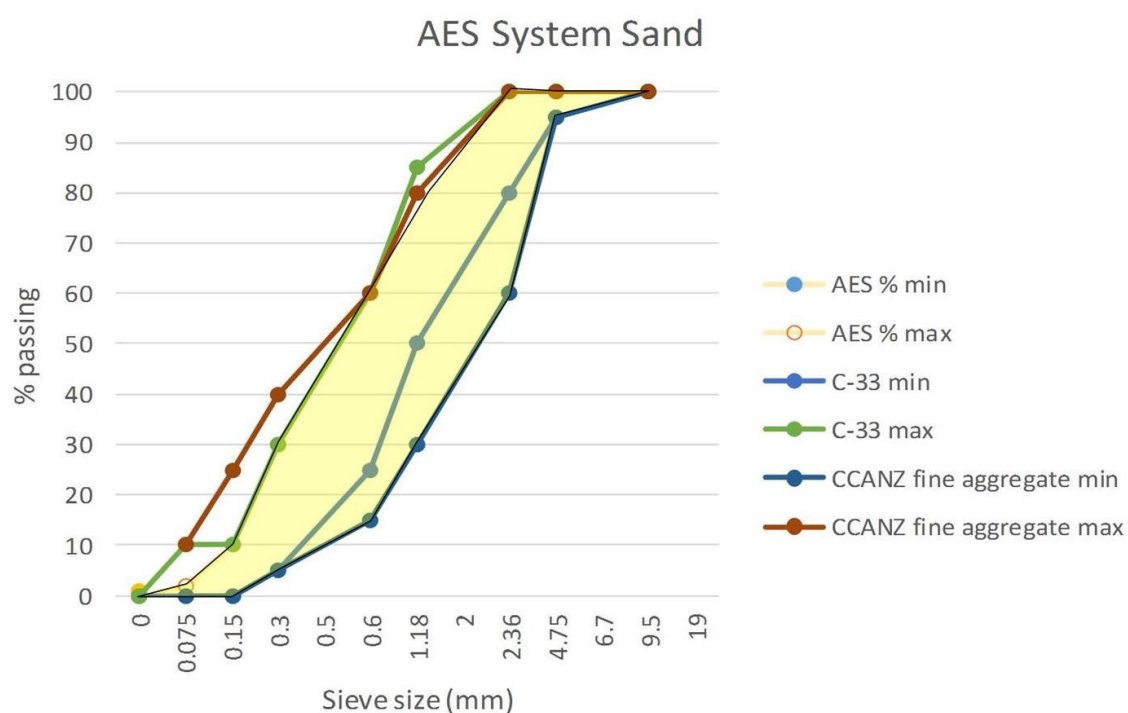
Environment Technology Ltd has sought to bring clarification to sand suppliers around New Zealand with regards to common sand specifications and the following graph details these sands. The two main sand specifications used are C33 and the CCANZ fine aggregates.

The top orange line of C33 sand (maximum percentage passing through sieve) with the exception of sands below 0.15mm and the bottom dark blue line of the CCANZ sand (minimum percentage passing through sieve) give an 'envelope' which is suitable for use as AES system sand. In Pap7 sand in Northland can be used as AES System Sand.

The target particle size is between 0.5mm and 2.0mm. 100 percent of the sand sample could be within this range but it is not shown on the graph.

AES System Sand envelope - Sieve Size Percent Passing (by weight)

9.5 mm (3/8 inch)	100
4.75 mm (No. 4)	95 to 100
2.36 mm (No. 8)	80 to 100
1.18 mm (No. 16)	50 to 85
600 µm (No. 30)	25 to 60
300 µm (No. 50)	5 to 30
150 µm (No. 100)	0 to 10
75 µm (No. 200)*	0 to 2

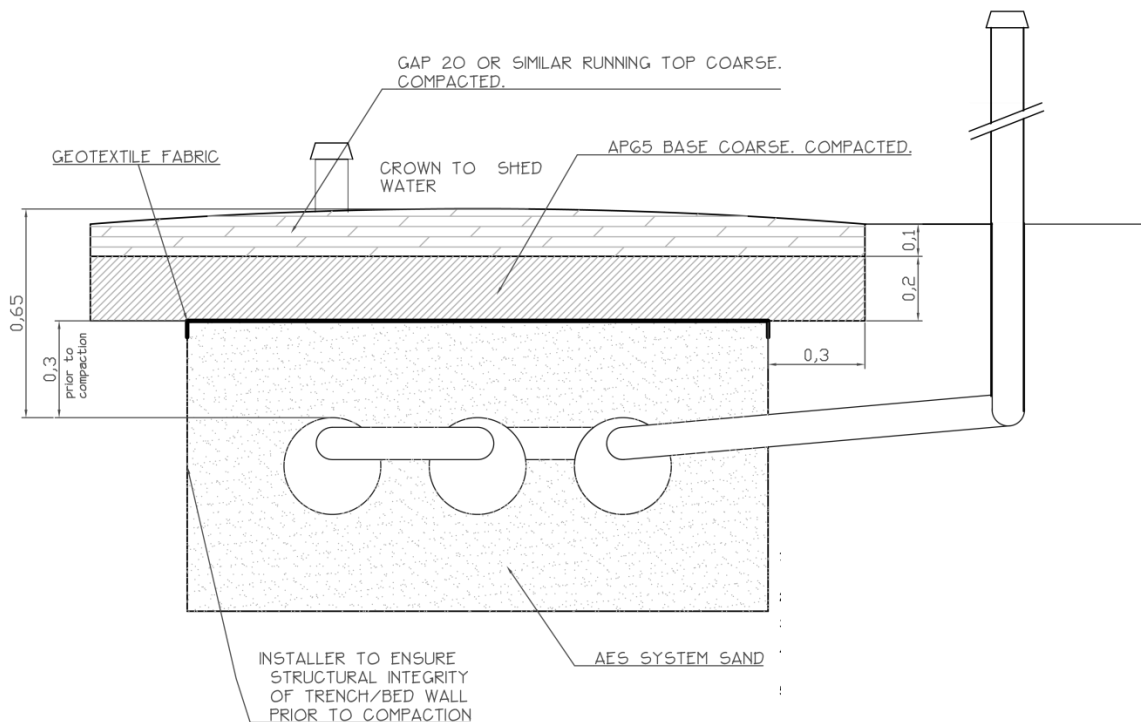


AES Traffic Loading Guidelines

AES technology has been used successfully under traffic-bearing surfaces for 20 years. AES pipe is strong and durable and can withstand vehicular loading with sufficient cover material. Due to the mandatory use of venting, systems installed under paved surfaces or hardscapes can function as effectively as those installed beneath soil.

The following information is being provided by Environment Technology Ltd. to use as a guideline for installation of the AES beds to traffic loading standards design for example when installed beneath drive-ways or car parks. For further clarification please contact Environment Technology at 03 970 7979 or 0800 WASTE H2O (0800 927 834).

Traffic Loading for AES beds



1. A minimum of 300mm structural cover (AP65) shall be compacted over 300mm of AES system sand. Structural cover material to extend at least 300mm beyond AES bed excavation to provide 'bridging'.
2. Surface cover may be GAP20 or similarly acceptable top coarse or local material free of organics, large stones and building debris.
3. Both structural cover layer and top coarse to be well compacted
4. AES system sand to be laid in 300mm layer then compacted to a uniform, dense stable condition prior to application of base coarse.
5. All base coarse and top coarse to be specified and applied as per Transit NZ TNZ B/02.2005 Specifications for Construction of Unbound Granular Pavement Layers.



Presby Environmental, Inc.

PRESBY ENVIRONMENTAL INC. STANDARD LIMITED WARRANTY

a) The structural integrity of each unit, endcap and other accessory manufactured by Presby Environmental Inc. (collectively referred to as "Units"), when installed and operated in an onsite wastewater system in accordance with Presby Environmental's installation instructions, is warranted to the original purchaser ("Holder") against defective materials and workmanship for one year from the date upon which a septic permit is issued for the septic system containing the Units; provided, however, that if a septic permit is not required for the septic system by applicable law, the one (1) year warranty period will begin upon the date that installation of the septic system commences. In order to exercise its warranty rights, Holder must notify Presby Environmental in writing at its corporate headquarters in Whitefield, New Hampshire within fifteen (15) days of the alleged defect. Presby Environmental will supply replacement Units for those Units determined by Presby Environmental to be defective and covered by this Limited Warranty. Presby Environmental's liability specifically excludes the cost of removal and/or installation of the Units.

(b) THE LIMITED WARRANTY AND REMEDIES IN SUBPARAGRAPH (a) ARE EXCLUSIVE. THERE ARE NO OTHER WARRANTIES WITH RESPECT TO THE UNITS, INCLUDING NO IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

(c) This Limited Warranty shall be void if any part of the Presby Environmental system (unit, endcap or other accessory) is manufactured by anyone other than Presby Environmental. The Limited Warranty does not extend to incidental, consequential, special or indirect damages. Presby Environmental shall not be liable for penalties or liquidated damages, including loss of production and profits, labor and materials, overhead costs, or other losses or expenses incurred by the Holder or any third party. Specifically excluded from Limited Warranty coverage are damage to the Units due to ordinary wear and tear, alteration, accident, misuse, abuse or neglect of the Units; the Units being subjected to vehicle traffic or other conditions which are not permitted by the installation instructions; failure to maintain the minimum ground covers set forth in the installation instructions; the placement of improper materials into the system containing the Units; failure of the Units or the septic system due to improper siting or improper sizing, excessive water usage, improper grease disposal, or improper operation; or any other event not caused by Presby Environmental. This Limited Warranty shall be void if the Holder fails to comply with all of the terms set forth in this Limited Warranty.

Further, in no event shall Presby Environmental be responsible for any loss or damage to the Holder, the Units, or any third party resulting from installation or shipment, or from any product liability claims of Holder or any third party. For this Limited Warranty to apply, the Units must be installed in accordance with all site conditions required by state and local codes; all other applicable laws; and Presby Environmental's installation instructions.

(d) No representative of Presby Environmental has the authority to change this Limited Warranty in any manner whatsoever, or to extend this Limited Warranty. No warranty applies to any party other than the original Holder.

The above represents the standard Limited Warranty offered by Presby Environmental. A limited number of states and counties have different warranty requirements. Any purchaser of Units should contact Presby Environmental's corporate headquarters in Whitefield, New Hampshire, prior to such purchase, to obtain a copy of the applicable warranty, and should carefully read that warranty prior to the purchase of Units.

PRESBY ENVIRONMENTAL, INC.

143 Airport Road, Whitefield NH, 03598 • (800) 473-5298 • www.presbyeco.com